

Data Dependency Analysis — EMS Optimization Project

Generated: 2026-03-15 | Scope: Full project data flow audit

1. Current State: Data Inventory

1.1 Raw Data (data/raw/)

File	Size	Git-Tracked	Source
Mo-tor_Vehicle_Collisions_-_Crashes_20260223.csv	536 MB	No (gitignored)	NYC Open Data download
Po-lice_Precincts_20260223.csv	3.6 MB	No (gitignored)	NYC Open Data download
FDNY_Firehouse_Listing_20260223.csv	36 KB	Yes	NYC Open Data download
manhattan_boundary.geojson	248 KB	Yes	Reference boundary
cbd_boundary.geojson	104 KB	Yes	MTA Congestion Relief Zone
nyc_borough_boundaries.geojson	3.1 MB	Yes	Reference boundary
manhattan_geom.pkl	100 KB	Yes	Generated (audit_step1)
cbd_geom.pkl	40 KB	Yes	Generated (audit_step1)
Mo-tor_Vehicle_Collisions_Data_Dictionary.xlsx	96 KB	Yes	Documentation
FDNY_Firehouse_Listing_Data_Dictionary.xlsx	68 KB	Yes	Documentation
Po-lice_Precincts_Data_Dictionary.xlsx	64 KB	Yes	Documentation

1.2 Processed Data (data/processed/)

File	Size	Git-Tracked	Generated By
crashes_manhattan.csv	99 MB	No	data_audit.py / audit_step4_crashes.py
crashes_manhattan.pqrquet	15 MB	No	data_audit.py / audit_step4_crashes.py
fire-houses_clean.csv	32 KB	Yes (force-added)	data_audit.py / audit_step2_firehouses.py
fire-houses_manhattan.csv	8 KB	Yes (force-added)	audit_step2_firehouses.py
pre-cincts_manhattan.geojson	640 KB	Yes (force-added)	data_audit.py / audit_step3_precincts.py
distance_matrix_firehouse_precinct.csv	28 KB	Yes (force-added)	run_optimization_comparison.py (via build_distance_matrix)
distance_matrix_firehouse_precinct_manhattan.csv	28 KB	Yes (force-added)	generate_manhattan_distance_matrix.py
demand_lambda_hourly.csv	4 KB	Yes (force-added)	demand_modeling.py
demand_lambda_dow.csv	4 KB	Yes (force-added)	demand_modeling.py
demand_lambda_precinct.csv	4 KB	Yes (force-added)	demand_modeling.py
demand_model_summary.json	4 KB	Yes (force-added)	demand_modeling.py

1.3 Results (results/)

417 files across subdirectories. All are generated outputs. None are git-tracked (gitignored). Generated by various scripts and notebooks.

1.4 Configs (configs/)

6 YAML files — all tracked in git. These are **reference/parameter** files, not generated.

2. Data Generation Chain (DAG)

```

TIER 0  External downloads (must exist before anything runs)
 Motor_Vehicle_Collisions_-_Crashes_20260223.csv [536 MB, NOT in repo]
 Police_Precincts_20260223.csv [3.6 MB, NOT in repo]
 FDNY_Firehouse_Listing_20260223.csv [36 KB, in repo]

TIER 0  Reference boundaries (in repo)
 manhattan_boundary.geojson
 cbd_boundary.geojson
 nyc_borough_boundaries.geojson

TIER 1  Geometry pickles (generated from Tier 0 boundaries)
 manhattan_geom.pkl  audit_step1_boundaries.py
 cbd_geom.pkl  audit_step1_boundaries.py

TIER 2  Spatial filtering (depends on Tier 0 + Tier 1)
 firehouses_clean.csv  audit_step2_firehouses.py
 firehouses_manhattan.csv  audit_step2_firehouses.py
 precincts_manhattan.geojson  audit_step3_precincts.py
 crashes_manhattan.csv  audit_step4_crashes.py
 crashes_manhattan.parquet  audit_step4_crashes.py

TIER 3  Demand modeling (depends on Tier 2)
 demand_lambda_hourly.csv  demand_modeling.py
 demand_lambda_dow.csv  demand_modeling.py
 demand_lambda_precinct.csv  demand_modeling.py
 demand_model_summary.json  demand_modeling.py

TIER 3  Distance matrices (depends on Tier 2)
 distance_matrix_firehouse_precinct.csv  run_optimization_comparison.py
 distance_matrix_firehouse_precinct_manhattan.csv  generate_manhattan_distance_matrix.py

TIER 4  Optimization results (depends on Tier 3)
 results/optimization/*

TIER 5  Simulation results (depends on Tier 3 + Tier 4)
 results/simulation/*

TIER 6  Statistical analysis & figures (depends on Tier 5)
 results/tables/*, results/figures/*, results/production_v2/*

```

3. Notebook → Data Dependencies

Notebook	Reads (Input)	Writes (Output)
02_eda_spatiotemporal	processed/ crashes_manhattan.parquet , processed/fire- houses_clean.csv , processed/pre- cincts_manhattan.geojson , raw/manhat- tan_boundary.geojson , raw/ cbd_boundary.geojson	results/figures/fig_*.png
03_input_modeling	processed/de- mand_lambda_hourly.csv , processed/de- mand_lambda_dow.csv , pro- cessed/de- mand_lambda_precinct.csv , processed/de- mand_model_summary.json	(visualization only)
04_service_travel_proxy	processed/dis- tance_matrix_firehouse_prec inct.csv	results/figures/*.png
05_optimization	results/optimization/ policy_comparison.csv , res- ults/optimization/sensitiv- ity_analysis.csv , results/ optimization/find- ings_summary.json , results/ optimization/alloca- tions_K40.csv	(visualization only)
06_simulation_debug	results/simulation/verific- ation/*.json , results/simu- lation/validation_pilot/ *.json	results/figures/*.png
07_production_results	results/simulation/produc- tion/exp1-4*.csv	results/figures/*.png , results/tables/ exp*_pivot_rt.csv
08_statistical_analysis	results/simulation/produc- tion/exp*.csv , results/ tables/table*.csv	(visualization only)
09_cbd_analysis		results/figures/*.png

Notebook	Reads (Input)	Writes (Output)
	results/simulation/ cbd_experiment/*.csv	
Colab Pipeline	All processed/*.csv , processed/*.geojson , configs/*.yaml	results/* (full pipeline)

4. Proposed Architecture Assessment

4.1 The Proposal

Only `raw/` and `reference/` data from the repo; all processed/intermediate data generated by notebooks; serialized artifacts for reuse; seed-controlled reproducibility.

4.2 Feasibility Verdict: Feasible with caveats

What works perfectly

- **Tier 1-3 data is fully reproducible.** The generation chain `raw → processed` is deterministic. Given the same raw CSV inputs and boundary GeoJSONs, the audit scripts + demand modeling will always produce identical outputs. No randomness is involved in spatial filtering, crash subsetting, or lambda table computation.
- **Distance matrices are deterministic.** Pure math from coordinates — no randomness.
- **SeedManager is already in place.** The project has a well-designed `SeedManager` class with per-component seed derivation, `configs/reproducibility.yaml`, and support for component overrides. Simulation results (Tier 4-6) are reproducible given the same seeds and processed inputs.

Caveats

1. **Two large raw files are NOT in the repo** (`Motor_Vehicle_Collisions_*.csv` at 536 MB, `Po-lice_Precincts_*.csv` at 3.6 MB). These are gitignored for good reason (size). A new user must download them separately. This is the **critical external dependency** — without these, nothing downstream can be generated.
2. **The *.pkl files in raw/ are misplaced.** `manhattan_geom.pkl` and `cbd_geom.pkl` are generated artifacts from `audit_step1_boundaries.py`, not raw data. They should either be in `processed/` or regenerated on the fly. Currently tracked in git — this is redundant since they're trivially derived from the tracked GeoJSONs.
3. **Processed files are currently git-tracked despite gitignore.** Nine processed files were force-added (`git add -f`). This creates a confusing state: gitignore says ignore them, but they're tracked. Under the proposed architecture, these should be removed from tracking.
4. **No single orchestrator script.** The generation chain requires running 6+ scripts in the correct order. There's no `make all` or `pipeline.py` that runs Tier 0 → Tier 6 end-to-end.
5. **data_processing.py referenced in Makefile doesn't exist.** The Makefile's `data` target calls `src/ems_readiness/data_processing.py` which is missing. The actual processing is done by the `audit_step*.py` scripts.

6. `crashes_manhattan.csv` (99 MB) is expensive to regenerate. Processing the 536 MB raw crashes file takes significant time. A user would need to do this once, then reuse the parquet version.

5. Gap Analysis

Current State	Proposed State	Gap	Effort
9 processed files tracked in git	0 processed files tracked	Remove from git tracking	Low
pkl files in <code>raw/</code>	pkl files generated or eliminated	Move generation to pipeline	Low
No orchestrator script	Single <code>make pipeline</code> or <code>run_all.py</code>	Create orchestrator	Medium
<code>data_processing.py</code> missing	Working Makefile target	Fix Makefile or create script	Medium
Raw CSVs must be manually downloaded	Documented download instructions	Create <code>data/raw/README.md</code>	Low
Notebooks read pre-existing results	Notebooks can run end-to-end	Already mostly works; needs ordering docs	Low
Colab notebook reads <code>processed/</code> directly	Colab has data generation cells	Already has setup notebook (<code>00_colab_setup_and_data</code>)	None

6. Recommendations

6.1 Immediate (Low effort, high impact)

1. Remove processed files from git tracking:

```
bash
git rm --cached data/processed/*.csv data/processed/*.json data/processed/*.geojson
git rm --cached data/processed/*.parquet # if any were tracked
git rm --cached data/raw/manhattan_geom.pkl data/raw/cbd_geom.pkl
git rm --cached data/manifests/data_audit_summary.csv
git commit -m "chore: stop tracking generated processed data"
```

2. Update `.gitignore` to also cover:

```
data/processed/*.json
```



```
data/processed/*.geojson
data/raw/*.pkl
data/manifests/*.csv
```

3. **Create** `data/raw/README.md` documenting how to obtain the two large CSV files (NYC Open Data URLs, expected filenames, expected row counts for verification).

6.2 Short-term (Medium effort)

1. **Create a pipeline orchestrator** (`scripts/run_pipeline.py` or enhance the Makefile):

```
Step 1: audit_step1_boundaries.py    → pkl files
Step 2: audit_step2_firehouses.py    → firehouses_clean/manhattan.csv
Step 3: audit_step3_precincts.py     → precincts_manhattan.geojson
Step 4: audit_step4_crashes.py       → crashes_manhattan.csv/parquet
Step 5: demand_modeling.py           → lambda tables
Step 6: (distance matrix generation) → distance_matrix*.csv
Step 7: run_optimization_comparison  → optimization results
Step 8: run_verification.py          → verification results
Step 9: run_validation_pilots.py      → validation results
Step 10: run_production_v2.py        → production results
```

2. **Fix the Makefile** — point the `data` target at the actual audit scripts or the new orchestrator.
3. **Eliminate the pkl intermediaries.** Have `audit_step2/3/4` read the GeoJSONs directly instead of requiring pre-pickled geometries. This removes a fragile intermediate step.

6.3 Long-term (Nice to have)

1. **Serialize expensive intermediate objects** (e.g., `crashes_manhattan.parquet`) with content hashes so the pipeline can skip steps when inputs haven't changed.
 2. **Add a `--verify` flag** to the pipeline that checksums all generated files against known-good values (for the default seed=42 run).
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7. Example End-to-End Workflow (Post-Migration)

```
# 1. Clone the repo
git clone <repo-url> && cd ems-optimization

# 2. Setup environment
make setup # or: pip install -r requirements.txt

# 3. Download external data (manual step – documented in data/raw/README.md)
#   Place these in data/raw/:
#       - Motor_Vehicle_Collisions_-_Crashes_20260223.csv
#       - Police_Precincts_20260223.csv

# 4. Generate ALL processed data from raw (deterministic, no seeds needed)
make data # or: python scripts/run_pipeline.py --step data

# 5. Run optimization (deterministic – LP solver, no randomness)
python scripts/run_optimization_comparison.py

# 6. Run simulation experiments (seed-controlled, reproducible)
python scripts/run_verification.py
python scripts/run_validation_pilots.py
python scripts/run_production_v2.py
#   → All use SeedManager(master_seed=42) from configs/reproducibility.yaml
#   → Identical results every time

# 7. Open notebooks to explore/visualize
jupyter lab notebooks/
```

Reproducibility Guarantee

Stage	Reproducibility Mechanism
Raw → Processed (Tiers 1-3)	Deterministic — no randomness involved
Optimization (Tier 4)	Deterministic — LP/MIP solver with fixed parameters
Simulation (Tier 5)	Seed-controlled — <code>SeedManager(master_seed=42)</code> with per-component derived seeds
Statistical Analysis (Tier 6)	Deterministic — operates on fixed simulation outputs

With `master_seed=42` (the default), any user running the full pipeline on the same raw data will get **bit-identical results** for Tiers 1-4 and **statistically identical results** for Tier 5 (same seeds → same random streams → same simulation outcomes).

8. File Classification Summary

Should be in repo (raw + reference)

- data/raw/FDNY_Firehouse_Listing_20260223.csv ✓
- data/raw/manhattan_boundary.geojson ✓
- data/raw/cbd_boundary.geojson ✓
- data/raw/nyc_borough_boundaries.geojson ✓
- data/raw/*_Data_Dictionary.xlsx (3 files) ✓
- configs/*.yaml (6 files) ✓

Should NOT be in repo (too large, must download)

- data/raw/Motor_Vehicle_Collisions_-_Crashes_20260223.csv (536 MB) ✓ already gitignored
- data/raw/Police_Precincts_20260223.csv (3.6 MB) ✓ already gitignored

Should NOT be in repo (generated, currently force-tracked — needs fixing)

- data/processed/firehouses_clean.csv
- data/processed/firehouses_manhattan.csv
- data/processed/precincts_manhattan.geojson
- data/processed/distance_matrix_firehouse_precinct.csv
- data/processed/distance_matrix_firehouse_precinct_manhattan.csv
- data/processed/demand_lambda_hourly.csv
- data/processed/demand_lambda_dow.csv
- data/processed/demand_lambda_precinct.csv
- data/processed/demand_model_summary.json
- data/raw/manhattan_geom.pkl (generated intermediate)
- data/raw/cbd_geom.pkl (generated intermediate)
- data/manifests/data_audit_summary.csv

Generated outputs (correctly not tracked)

- data/processed/crashes_manhattan.csv ✓
- data/processed/crashes_manhattan.parquet ✓
- results/**/* (417 files) ✓